MODELOS DE DEEP LEARNING PARA PREDECIR FUNCIONES DE PROTEÍNAS

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Universidad Nacional de Entre Ríos
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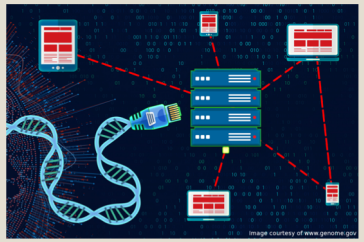
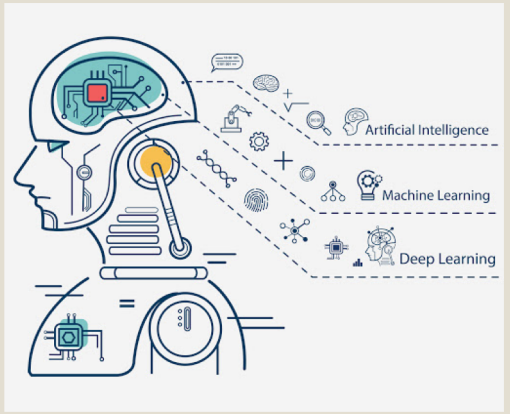
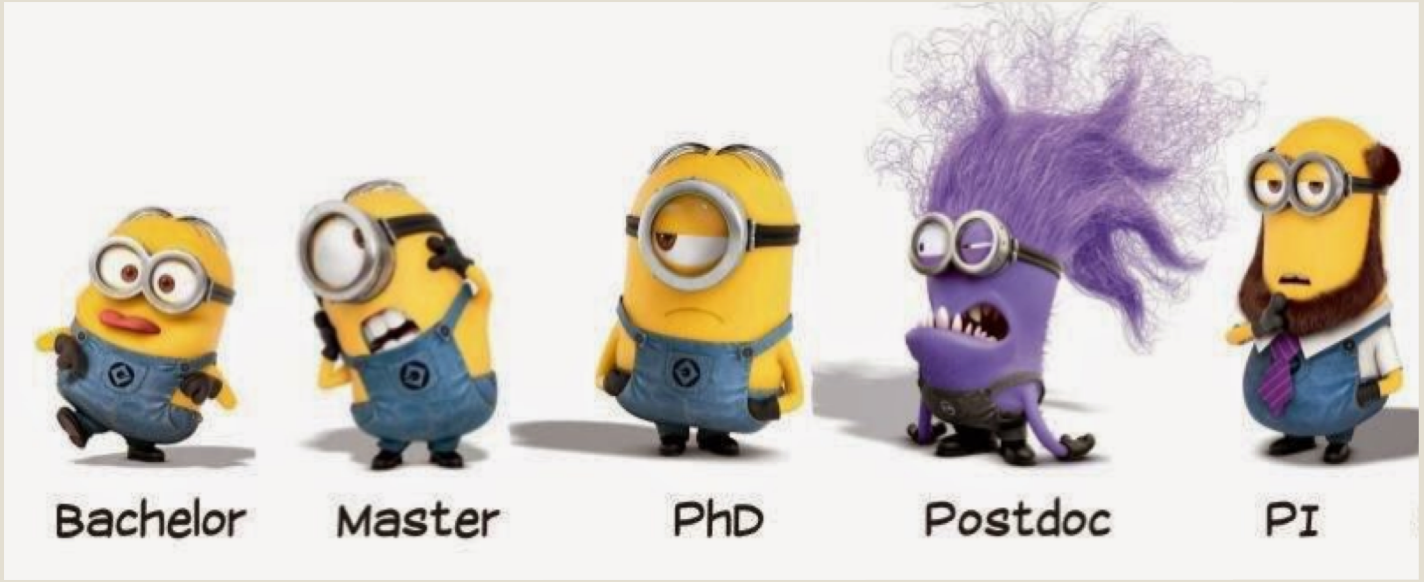
s i n c (*i*)

Sobre mi ...




Artificial Intelligence
Machine Learning
Deep Learning


Bachelor
Master
PhD
Postdoc
PI

Sobre mi …

"Desarrollo de algoritmos inteligentes para inferior funciones de proteínas a partir de la integración de datos"

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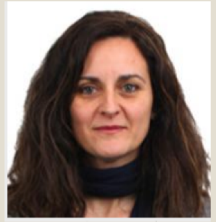

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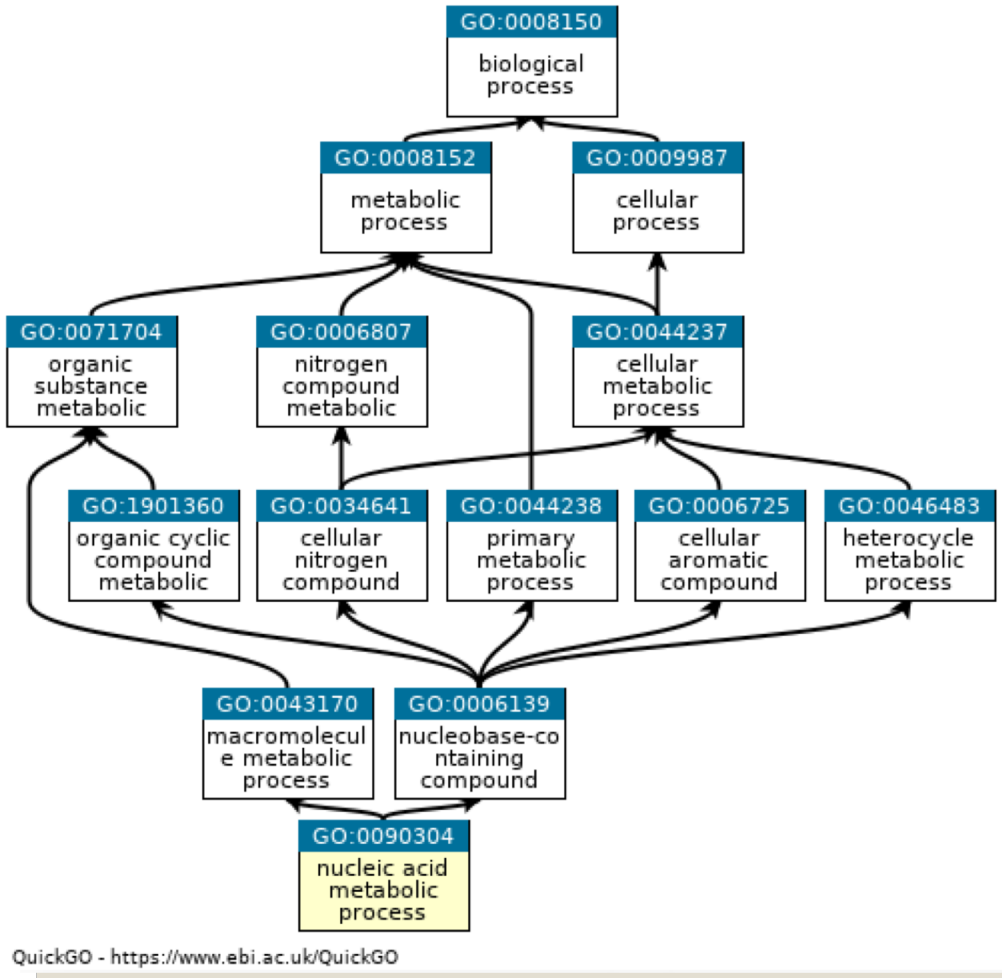
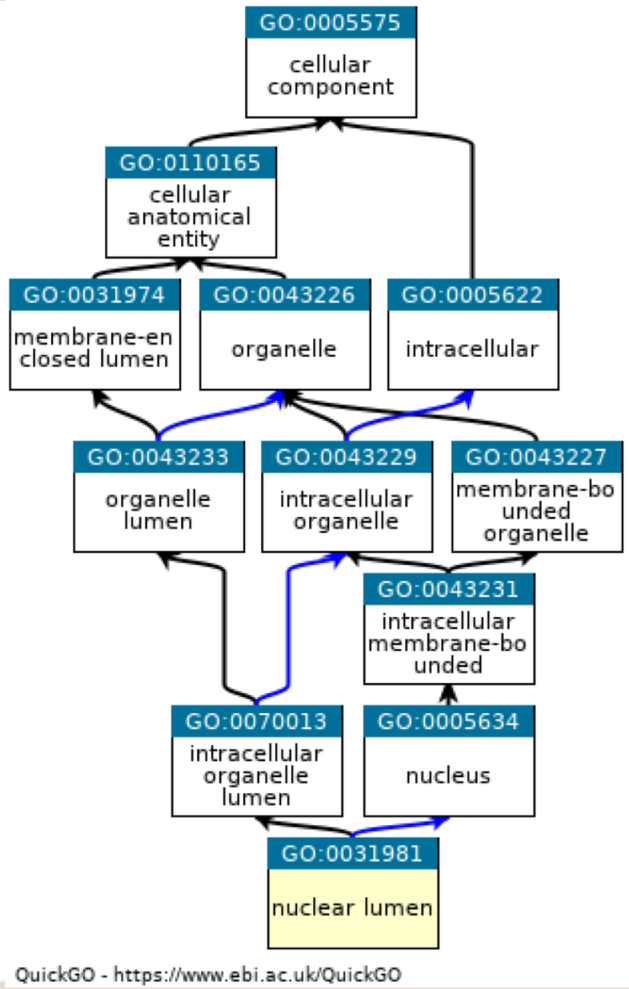
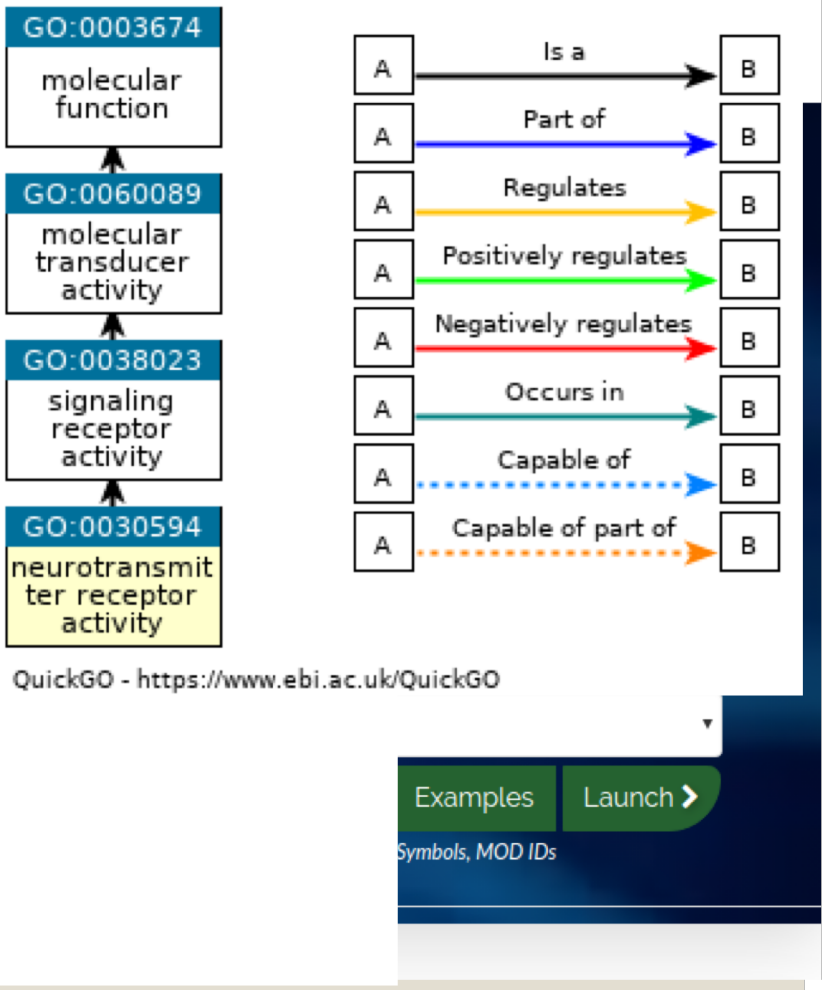
¿Cómo están representadas las funciones de proteínas?







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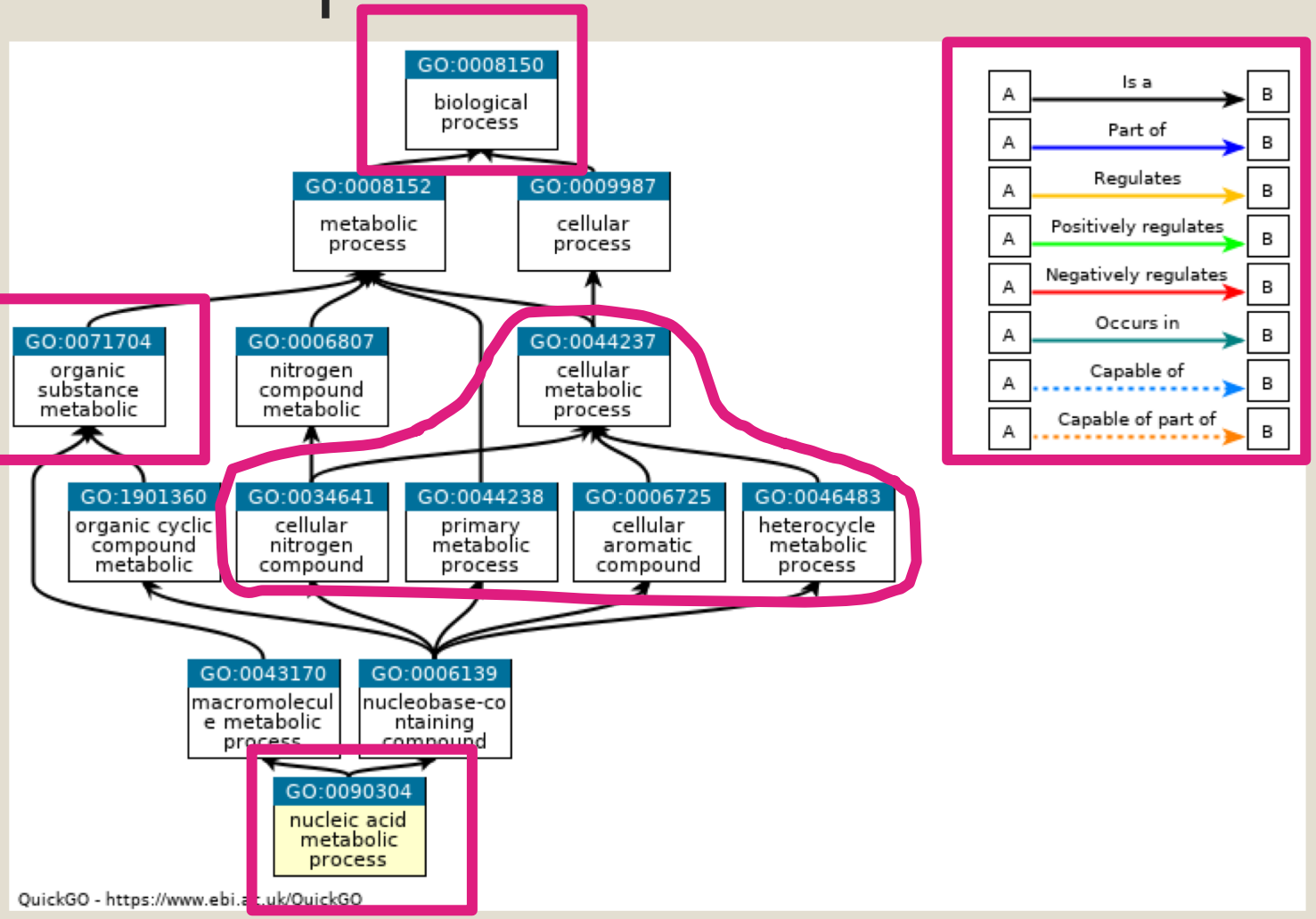
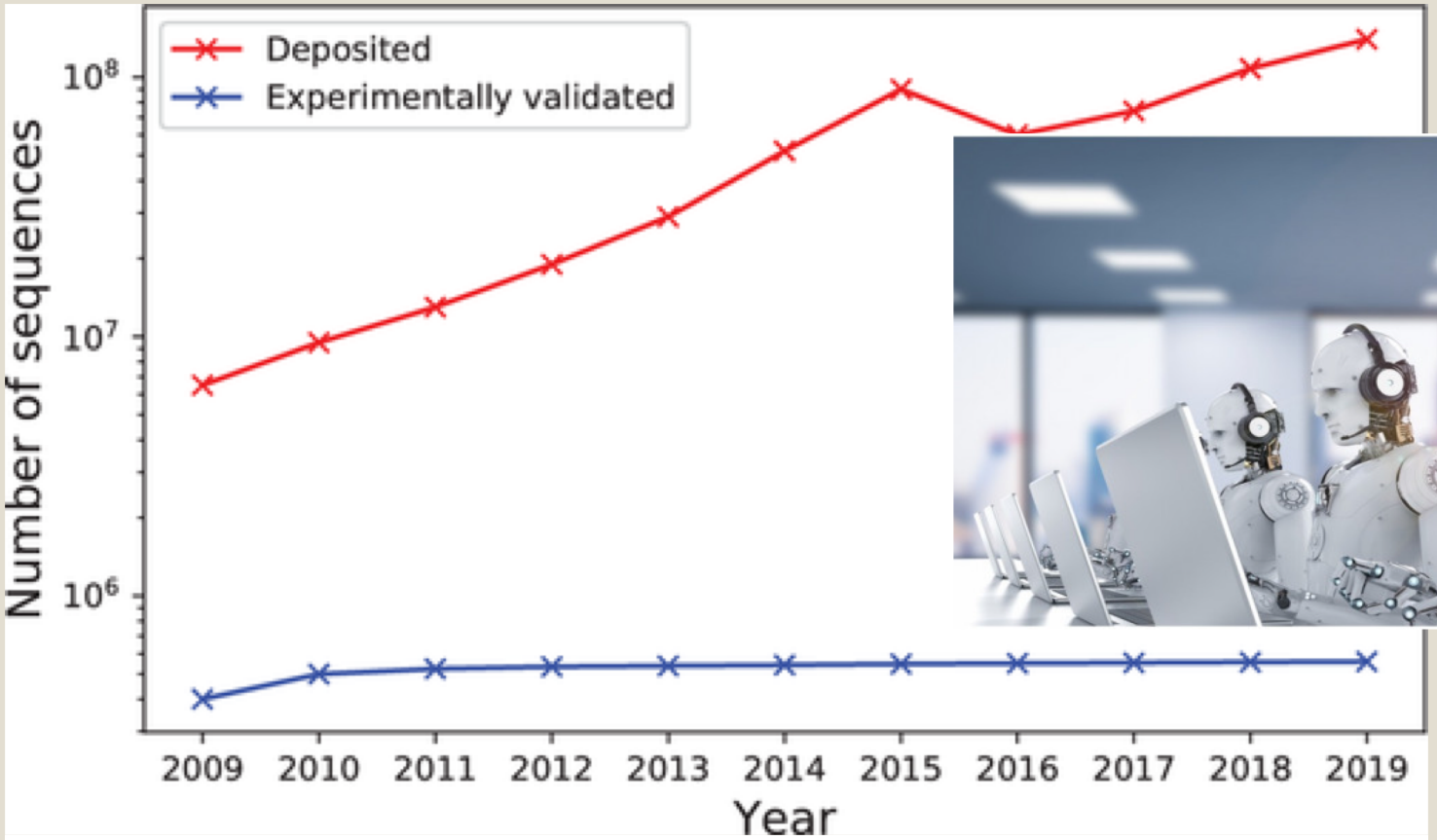


¿Cómo están representadas las funciones de proteínas?

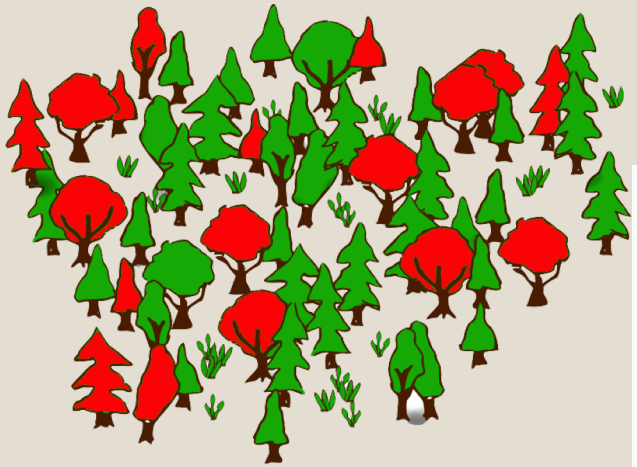
Gene Product	Symbol	Qualifier	GO Term	Evidence	Reference	With / From	Assigned By
UniProtKB:P09368	PUT1	enables	GO:0071949 (F) FAD binding	ECO:0000318 IBA	PMID:21873635	PANTHER:PTN000357758 more...	GO_Central
UniProtKB:P09368	PUT1	enables	GO:0004657 (F) proline dehydrogenase activity	ECO:0000318 IBA	PMID:21873635	FB:FBgn0003423 more...	GO_Central
UniProtKB:P09368	PUT1	part_of	GO:0005739 (C) mitochondrion	ECO:0000318 IBA	PMID:21873635	FB:FBgn0003423 more...	GO_Central
UniProtKB:P09368	PUT1	involved_in	GO:0010133 (P) proline catabolic process to glutamate	ECO:0000318 IBA	PMID:21873635	PANTHER:PTN000357758 more...	GO_Central
UniProtKB:P09368	PUT1	involved_in	GO:0055114 (P) oxidation-reduction process	ECO:0000322 IEA	GO_REF:0000043	UniProtKB-KW:KW-0560	UniProt
UniProtKB:P09368	PUT1	involved_in	GO:0006560 (P) proline metabolic process	ECO:0000322 IEA	GO_REF:0000043	UniProtKB-KW:KW-0642	UniProt
UniProtKB:P09368	PUT1	part_of	GO:0005739 (C) mitochondrion	ECO:0000322 IEA	GO_REF:0000043	UniProtKB-KW:KW-0496	UniProt
UniProtKB:P09368	PUT1	enables	GO:0016491 (F) oxidoreductase activity	ECO:0000322 IEA	GO_REF:0000043	UniProtKB-KW:KW-0560	UniProt
UniProtKB:P09368	PUT1	enables	GO:0004657 (F) proline dehydrogenase activity	ECO:0000314 IDA	PMID:20450881		SGD
UniProtKB:P09368	PUT1	involved_in	GO:0006537 (P) glutamate biosynthetic process	ECO:0000305 IC	PMID:387737	GO:0010133	SGD
UniProtKB:P09368	PUT1	part_of	GO:0005739 (C) mitochondrion	ECO:0000314 IDA	PMID:3125423		SGD

¿Por qué la predicción automática?

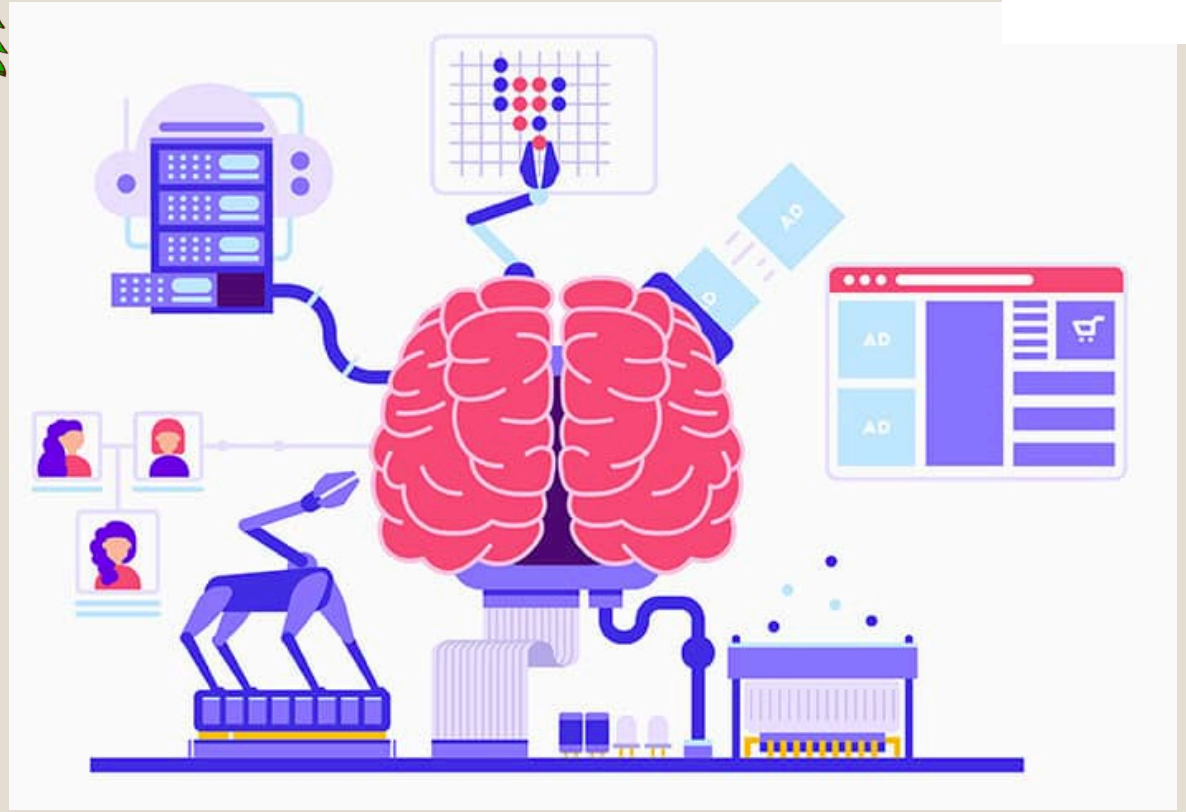


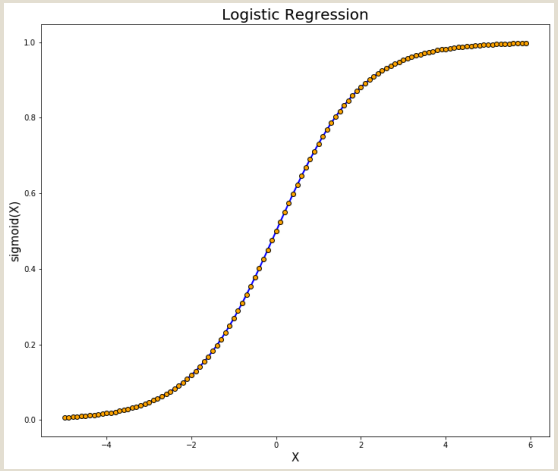
Number of sequences deposited and experimentally validated in UniProtKB over the past decade. The drop observed between 2015 and 2016 is due to procedures deployed by curators to identify and remove redundant proteomes. Figure extracted from Bonetta, R., & Valentino, G. (2020). Machine learning techniques for protein function prediction. *Proteins: Structure, Function, and Bioinformatics*, *88*(3), 397-413.

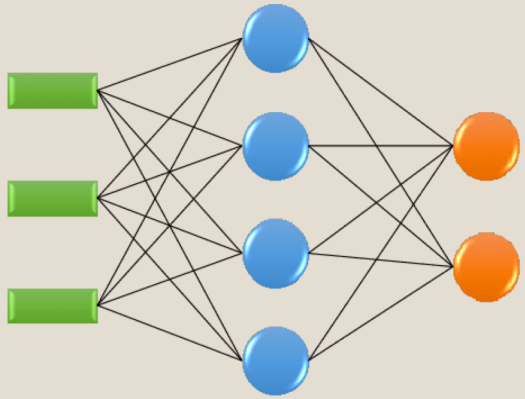
Estrategias para predecir funciones de proteínas



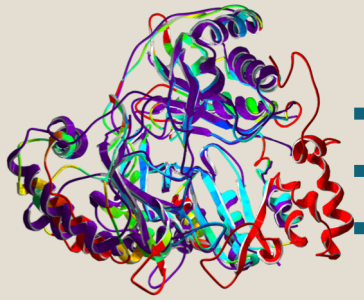
$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$







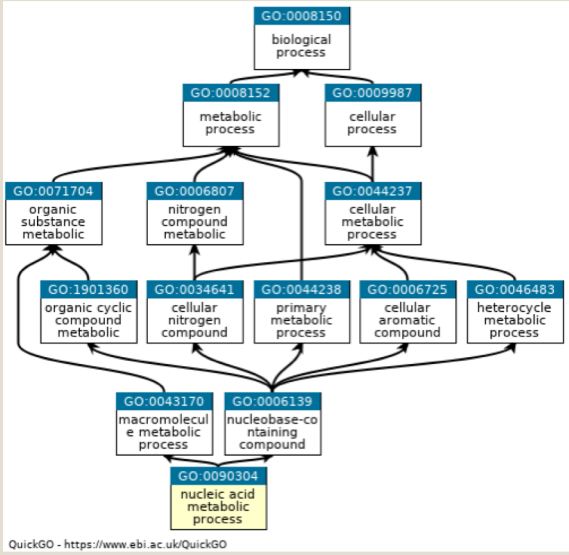
Estrategias para predecir funciones de proteínas



Feature extraction

Prediction model

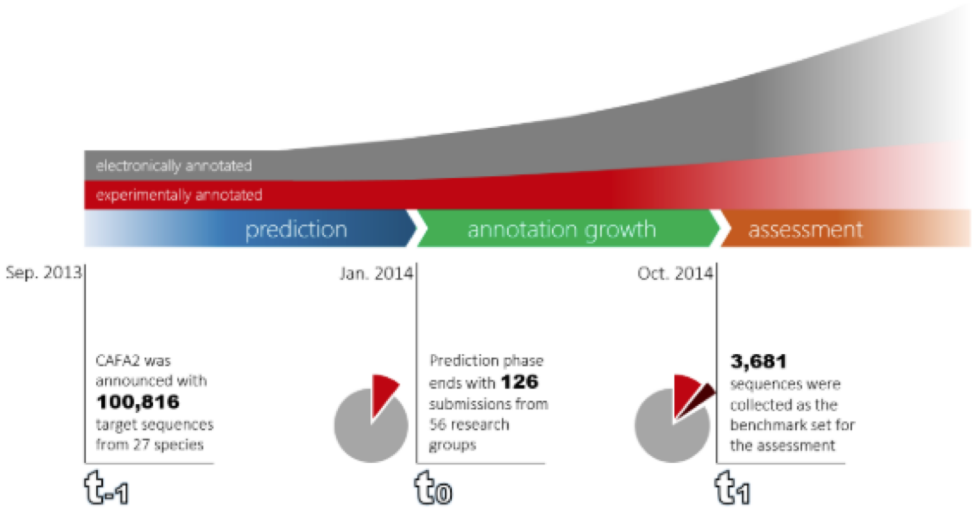
0.95
0.23
0.13
0.74
...
...
...
...
0.03
0.58
0.99
0.18



Estado del arte

The CAFA Challenge:

***The problem*:** There are many proteins in the databases for which the sequence is known, but the function is not. The gap between what we know and what we do not know is growing. A major challenge in the field of bioinformatics is to predict the function of a protein from its sequence or structure. At the same time, how can we judge how well these function prediction algorithms are performing?



***The solution*:** The Critical Assessment of protein Function Annotation algorithms (CAFA) is an experiment designed to provide a large-scale assessment of computational methods dedicated to predicting protein function, using a time challenge. Briefly, CAFA organizers provide a large number of protein sequences. The predictors then predict the function of these proteins by associating them with Gene Ontology terms or Human Phenoytpe Ontology terms (Blue "prediction" section of timeline). Following the prediction deadline, we wait for several months. During that time, some proteins whose function were unknown experimentally have received experimental verification (Green "annotation growth" section of timeline). Those proteins constitute the benchmark, against which the methods are tested (Orange "assessment" portion of timeline).

Genome Biology

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The CAFA challenge reports improved protein function prediction and new functional annotations for hundreds of genes through experimental screens

Naihui Zhou, Yuxiang Jiang, [...] Iddo Friedberg

Genome Biology **20**, Article number: 244 (2019) | Cite this article

Estado del arte

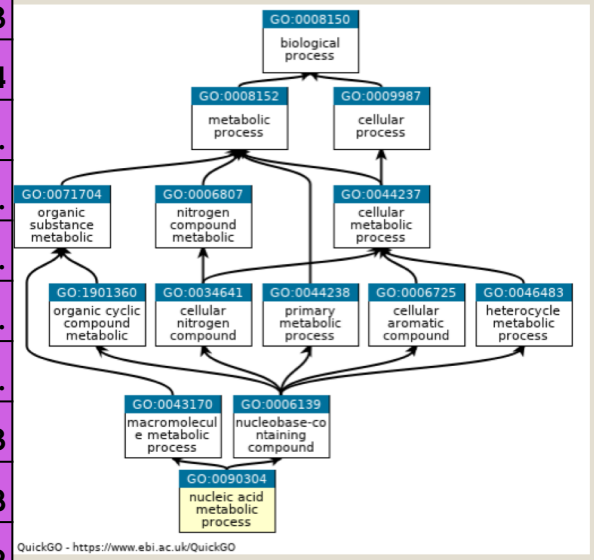
$$AvgRec(t) = \frac{1}{L}\sum_{i=1}^{L} recall_i(t)$$

$$AvgPre(t) = \frac{1}{v(t)}\sum_{i=1}^{v(t)} precision_i(t)$$

$$F - max = \max_t \left\{ \frac{2 * AvgPre(t) * AvgRec(t)}{AvgPre(t) + AvgRec(t)} \right\}$$

$i = i - th\ protein$
$t = threshold$
$f = f - th\ GO\ term$
$L = \#\ of\ annotated\ proteins$
$v(t) = \#\ of\ proteins\ with\ at\ least\ one\ GO\ term\ predicted\ at\ theshold\ t$

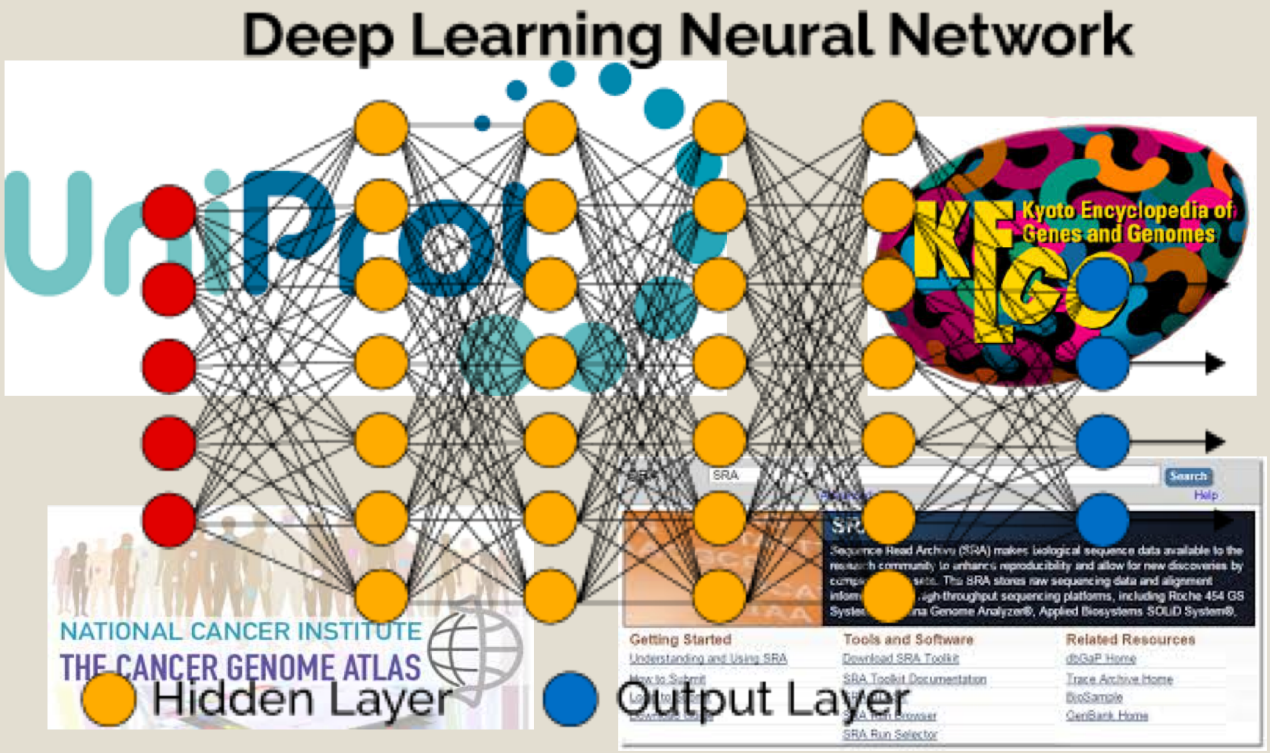
0.95
0.23
0.13
0.74
...
...
...
...
...
0.03
0.58
0.99
0.18



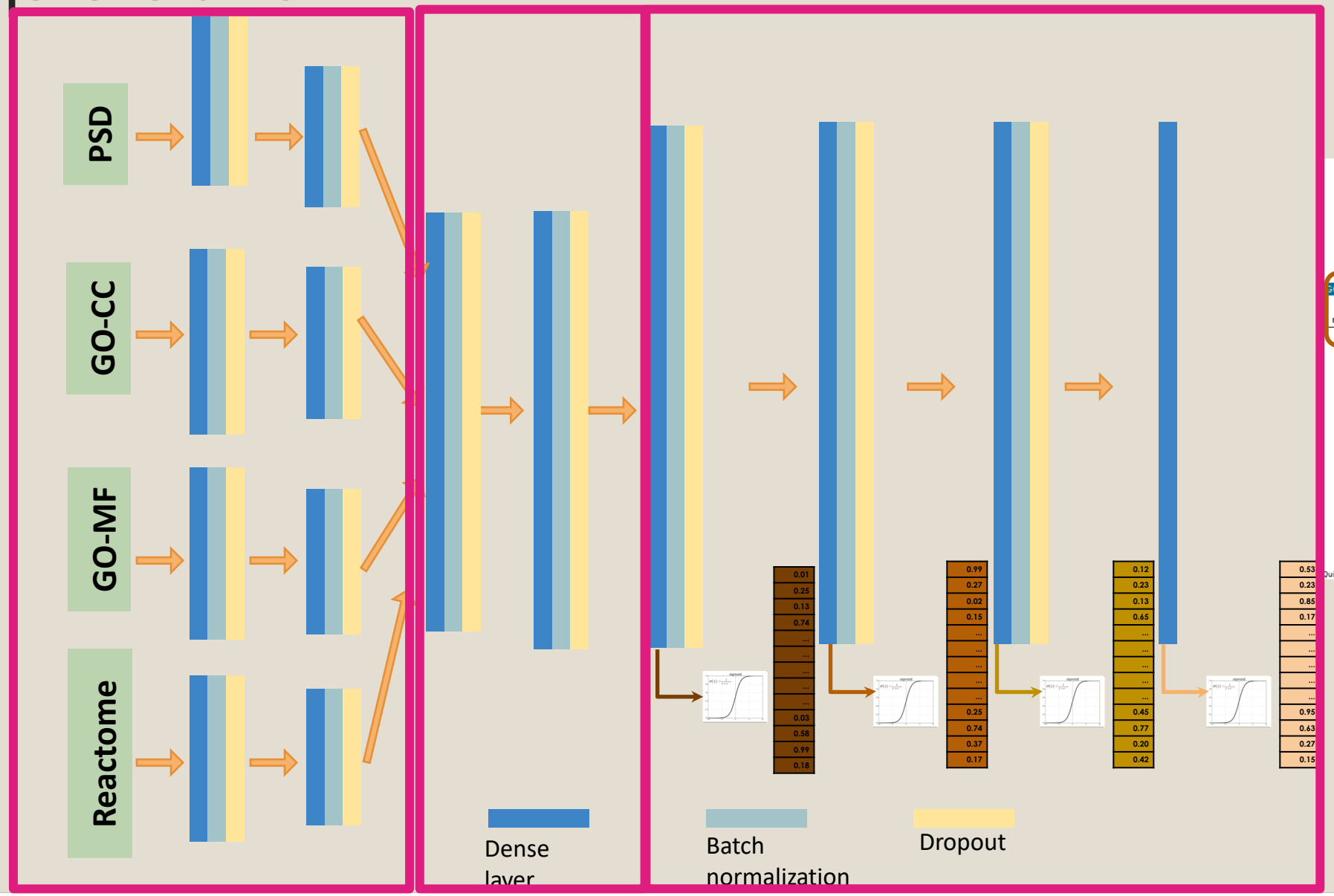
CLARK, Wyatt T.; RADIVOJAC, Predrag. Information-theoretic evaluation of predicted ontological annotations. *Bioinformatics*, 2013, vol. 29, no 13, p. i53-i61.

Propuesta

- Utilizar el conocimiento actual de las proteínas
- Reducir/evitar la extracción de características



Propuesta


PSD
GO-CC
GO-MF
Reactome
Dense layer
Batch normalization
Dropout
GO-BP

Resultados preliminares GO-BP

Datos:

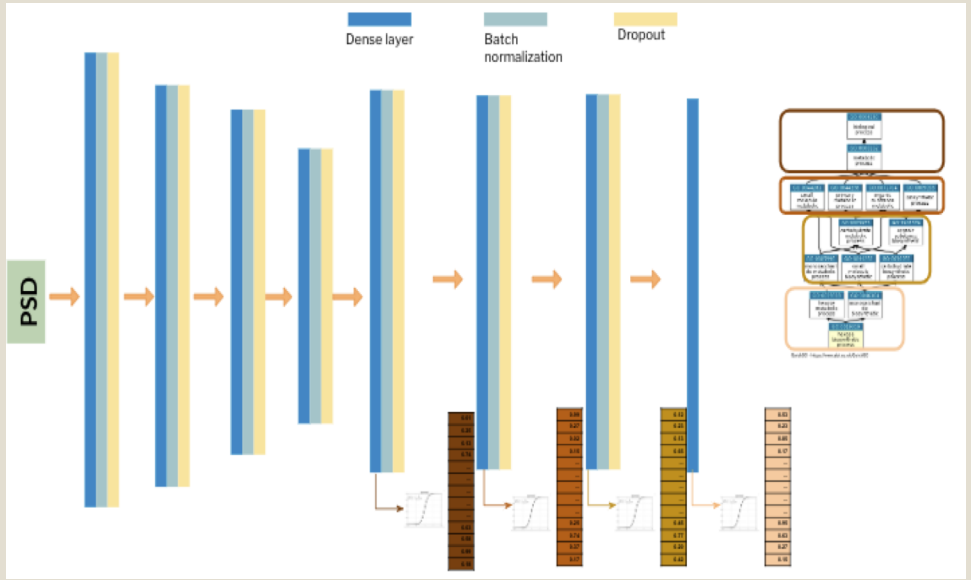
- 5.799 proteínas de S. cerevisiae
    - 3,360 con GO-BP
    - 4,200 con GO-CC
    - 2,613 con GO-MF
    - 1,459 con Reactome

Setup experimental

- 3-fold cross validation (70-10-20)
- Especificaciones técnicas
    - Early stopping y BS=16
    - Función de activación ELU
    - Dropout (0.25)
    - BCE loss
    - ADAM optimizer

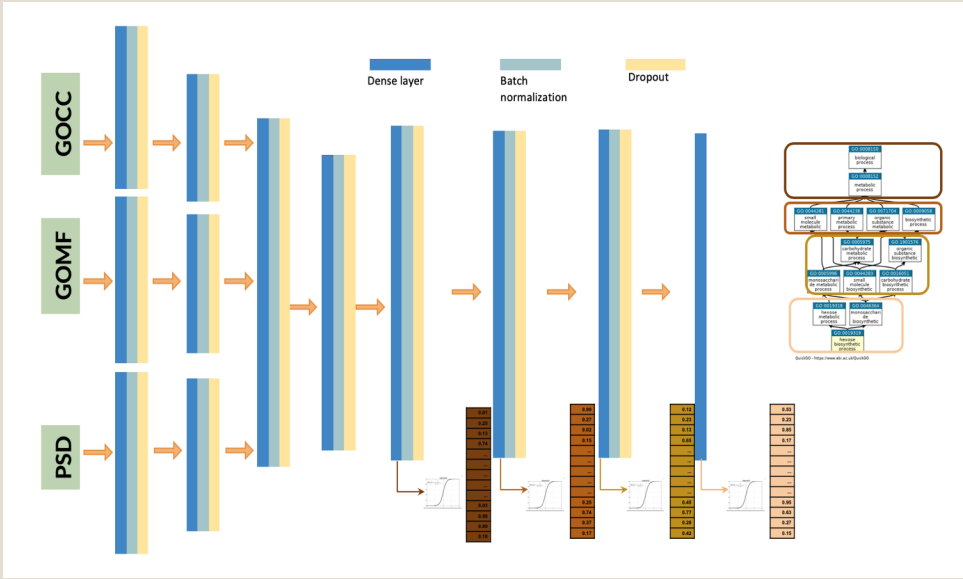
Resultados preliminares GO-BP

Input	F-max	Precision	Recall	Threshold
PSD	0.345	0.325	0.372	0.100



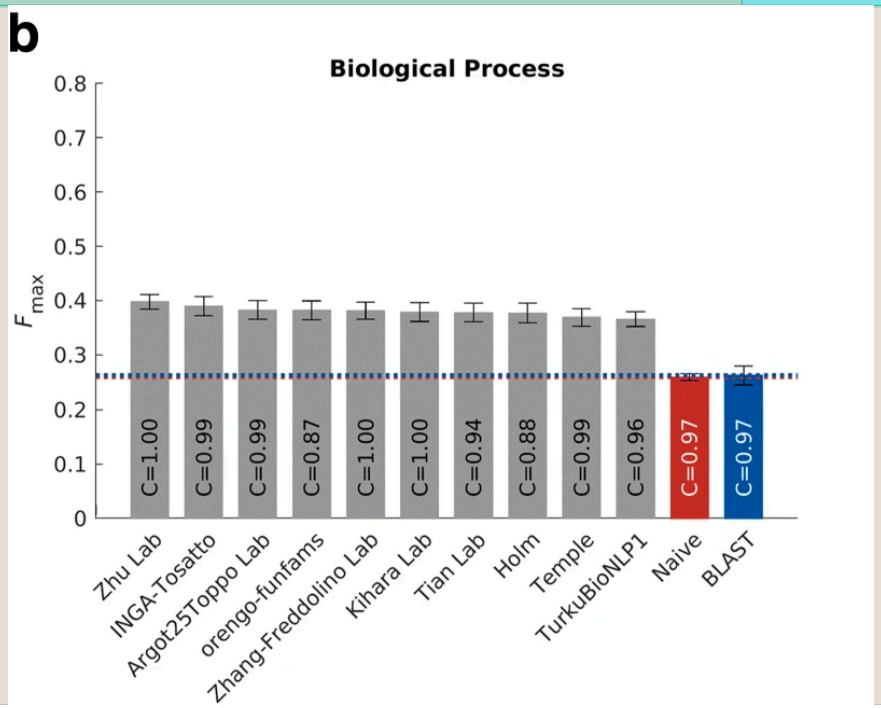
Resultados preliminares GO-BP

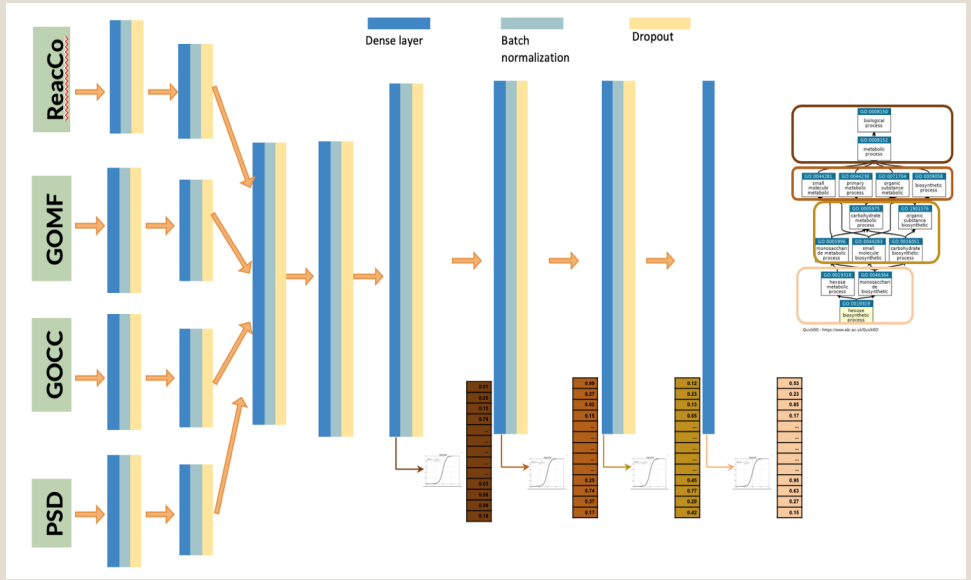
Input	F-max	Precision	Recall	Threshold
PSD	0.345	0.325	0.372	0.100
PSD-GOCC-GOMF	**0.467**	**0.519**	**0.426**	**0.233**



Resultados preliminares GO-BP

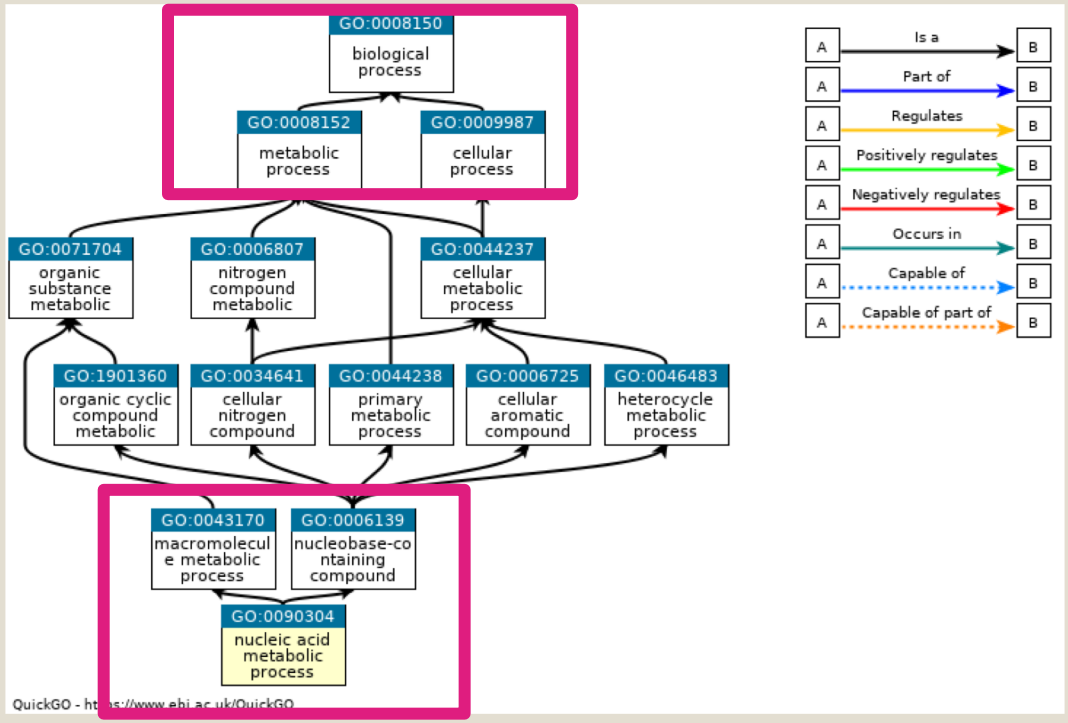
Input	F-max	Precision	Recall	Threshold
PSD	0.345	0.325	0.372	0.100
PSD-GOCC-GOMF	0.467	0.519	0.426	0.233
PSD-GOCC-GOMF-Reactome	**0.480**	0.518	**0.452**	0.200





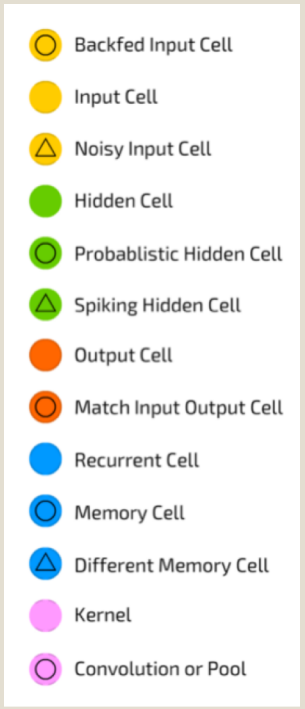
Trabajo futuro

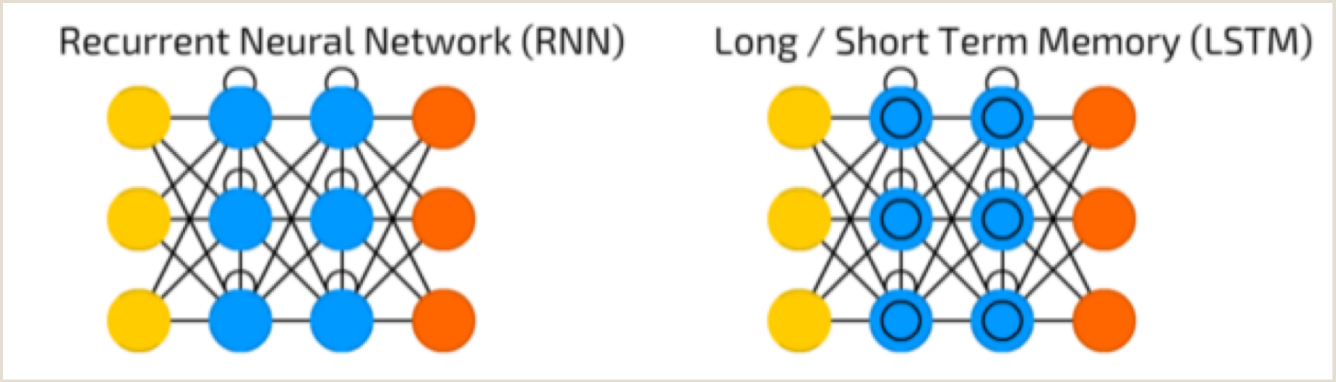
- Analizar qué tan bien representados están los GO terms en los datos de training

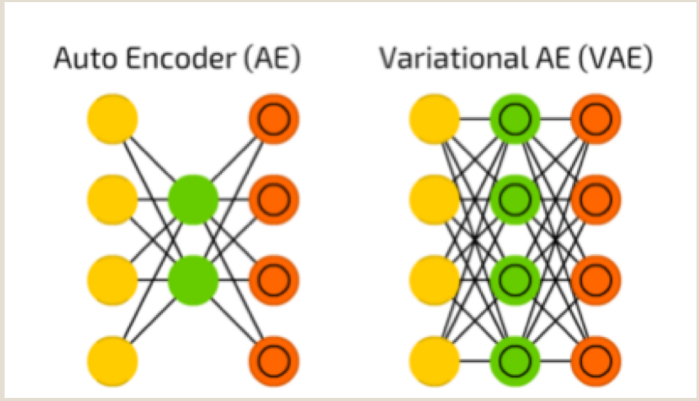


Trabajo futuro

- Probar otras arquitecturas de DL







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Diego Milone

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Víctor H. Casco

María J. Martin

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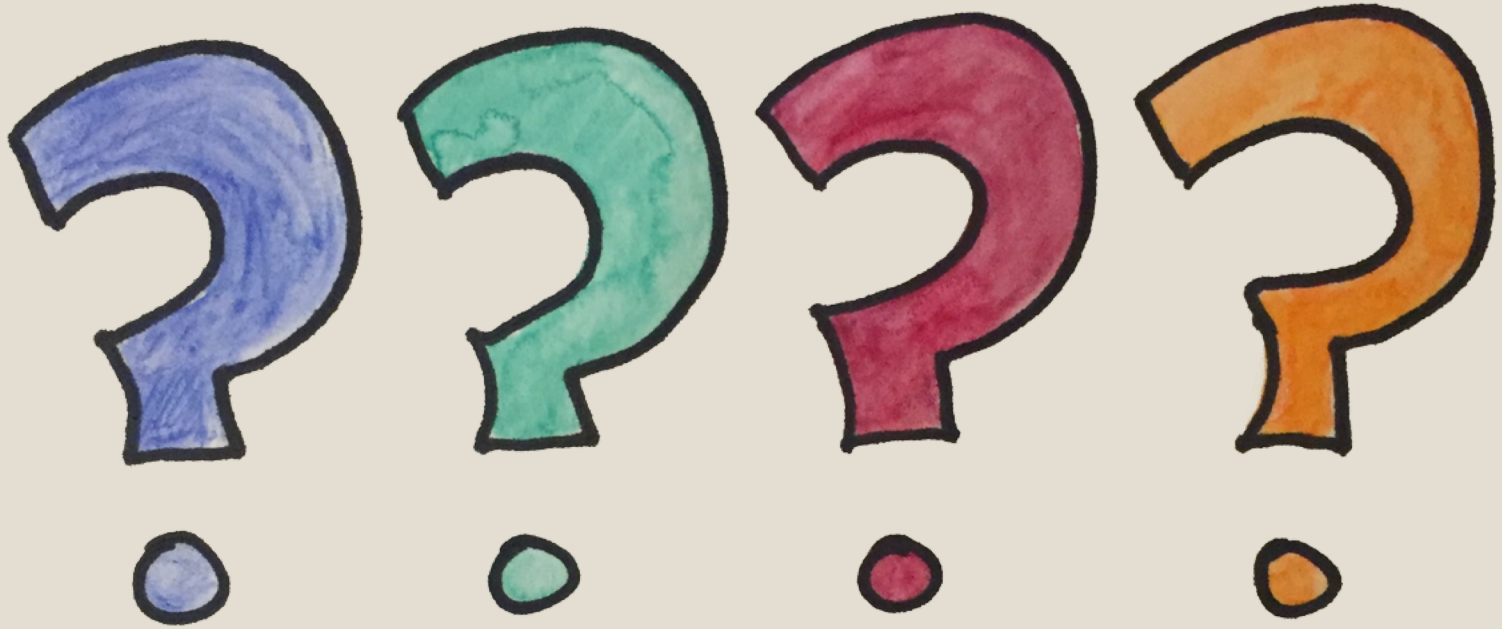
‘The strongest actions for a woman is to *love herself*, *be herself* and *shine* amongst those who never believed *she could*’



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